

- Q.29 Compare properties of ring spun, rotor-spun, air-jet spun and friction spun yarns.
- Q.30 What is Drawing? Why it is done in two stages?
- Q.31 What is the function of the transport channel in rotor spinning?
- Q.32 Compare the principle of yarn formation in rotor spinning with ring spinning.
- Q.33 Explain fiber dyeing and its importance in yarn manufacturing.
- Q.34 Write a detailed note on fibre dyeing antistatic finish and segregation.
- Q.35 Explain the texturing process and its importance in synthetic yarn manufacturing.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Explain the basic elements and principle of the rotor spinning machine. Describe in detail the passage of fibres through the rotor spinning frame with a neat diagram.
- Q.37 Explain the principle and working of friction (DREF) spinning. Describe the process of yarn formation with a neat diagram.
- Q.38 What is texturing? Discuss the principle, methods and importance of texturing in yarn manufacturing.

No. of Printed Pages : 4

182762/122762/032762

Roll No.

6th Sem / Textile Tech.

**Sub : Advanced Yarn Manufacturing/
Modern Methods in Yarn Product**

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Which of the following is a major limitation of ring spinning?
- Low yarn strength
 - High production rate
 - Balloon formation and yarn tension
 - High air consumption
- Q.2 Which fibre property most affects yarn strength?
- Length
 - Colour
 - Density
 - Softness
- Q.3 Which part of the machine produces yarn twist in air jet spinning?
- Rotor
 - Ring traveller
 - Air nozzles
 - Transport channel
- Q.4 Friction spinning is also called:
- Rotor spinning
 - Draft spinning
 - Self twist spinning
 - Air vortex spinning
- Q.5 The object of heat setting is to:
- Lock molecular orientation & set crimp
 - Increase fibre fineness

- c) Reducer yarn twist
- d) Improve fibre length
- Q.6 Which is NOT a post spinning operations?
 - a) Drawing b) Heat setting
 - c) Texturing d) Ginning
- Q.7 Antistatic finishes are especially useful for:
 - a) Cotton and wool b) Polyester and nylon
 - c) Jute and flax d) Silk
- Q.8 Air Jet yarn is generally:
 - a) High strength
 - b) Smooth with low hairiness
 - c) Very coarse
 - d) Identical to ring yarn
- Q.9 Which yarn has the highest strength?
 - a) Ring yarn b) Rotor yarn
 - c) Air jet yarn d) Friction yarn
- Q.10 Production rate is highest in:
 - a) Ring spinning b) Rotor spinning
 - c) Air jet spinning d) Friction spinning

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Why is ring spinning slower than other spinning methods?
- Q.12 Compared to speed of rotor spinning, ring spinning is ____ (faster / Slower)
- Q.13 Is the twist in air jet spinning true or false?

(2) 182762/122762/032762

- Q.14 State one limitation of air jet spun yarn.
- Q.15 Name one limitation of friction spun yarn.
- Q.16 The cost of production in ring spinning is high due to low ____.
- Q.17 Name one advantage of fibre blending.
- Q.18 What is the main purpose of applying an antistatic finish?
- Q.19 What is false twist?
- Q.20 Name two open end spinning system.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Describe the major limitations of ring spinning.
- Q.22 Explain the effect of traveller wear (abrasion) and yarn ballooning on ring spinning performance.
- Q.23 Discuss the importance of fibre length, fineness and strength in rotor yarn.
- Q.24 What is the concept of Heat setting? Give the name of heat set fibres.
- Q.25 Explain the basic principle of yarn formation for open end spinning.
- Q.26 Explain the principle and working of air-jet spinning.
- Q.27 Describe the mechanism of yarn formation in friction spinning.
- Q.28 Discuss the advantages and limitations of friction spun yarn in comparison to other spinning systems.

(3) 182762/122762/032762